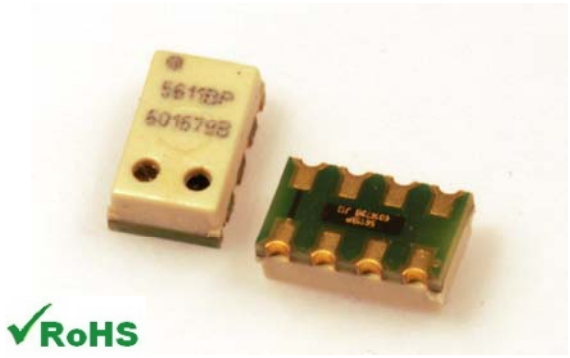
MS5611-01BA01 Variometer Module, with LCP cap

Not recommended for new designs



- **High resolution module, 10 cm**
- **Fast conversion down to 1 ms**
- **Low power, 1 µA (standby < 0.15 µA)**
- **QFN package 5.0 x 3.0 x 1.7 mm^3**
- **Supply voltage 1.8 to 3.6 V**
- **Integrated digital pressure sensor (24 bit ΔΣ ADC)**
- **Operating range: 10 to 1200 mbar, -40 to +85 °C**
- **I^{2}C and SPI interface up to 20 MHz**
- **No external components (Internal oscillator)**
- **Excellent long term stability**

DESCRIPTION

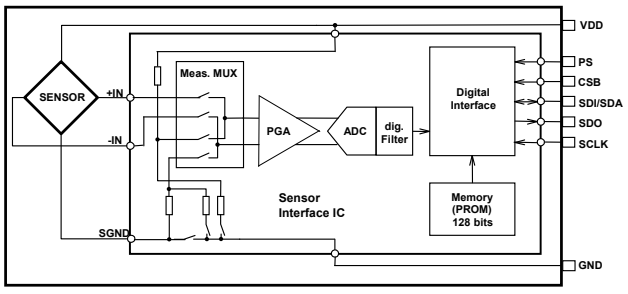
The MS5611-01BA is a new generation of high resolution altimeter sensors from MEAS Switzerland with SPI and I^{2}C bus interface. It is optimized for altimeters and variometers with an altitude resolution of 10 cm. The sensor module includes a high linearity pressure sensor and an ultra low power 24 bit ΔΣ ADC with internal factory calibrated coefficients. It provides a precise digital 24 Bit pressure and temperature value and different operation modes that allow the user to optimize for conversion speed and current consumption. A high resolution temperature output allows the implementation of an altimeter/thermometer function without any additional sensor. The MS5611-01BA can be interfaced to virtually any microcontroller. The communication protocol is simple, without the need of programming internal registers in the device. Small dimensions of only 5.0 mm x 3.0 mm and a height of 1.7 mm allow for integration in mobile devices. This new sensor module generation is based on leading MEMS technology and latest benefits from MEAS Switzerland proven experience and know-how in high volume manufacturing of altimeter modules, which have been widely used for over a decade. The sensing principle employed leads to very low hysteresis and high stability of both pressure and temperature signal.

FEATURES

FIELD OF APPLICATION

- Mobile altimeter / barometer systems
- Bike computers
- Variometers
- Height sensing for medical alarms
- Indoor navigation

FUNCTIONAL BLOCK DIAGRAM



TECHNICAL DATA

Sensor Performances (V_{DD} = 3 V)				
Pressure	**Min**	**Typ**	**Max**	**Unit**
Range	10		1200	mbar
ADC		24		bit
Resolution (1)		0.065 / 0.042 / 0.027 / 0.018 / 0.012		mbar
Accuracy 25°C, 750 mbar	-1.5		+1.5	mbar
Error band, -20°C to +85°C 450 to 1100 mbar (2)	-2.5		+2.5	mbar
Response time (1)		0.5 / 1.1 / 2.1 / 4.1 / 8.22		ms
Long term stability		-1		mbar/yr
Temperature	**Min**	**Typ**	**Max**	**Unit**
Range	-40		+85	°C
Resolution		<0.01		°C
Accuracy	-0.8		+0.8	°C

Notes: (1) Oversampling Ratio: 256 / 512 / 1024 / 2048 / 4096
(2) With autozero at one pressure point

MS5611-01BA01 带 LCP 盖的升降率计模块

原稿标注：不建议用于新设计。资料编号 DA5611-01BA01_006；日期 2011 年 7 月 19 日。

主要特点

高分辨率：高度分辨率 10 cm；最快约 1 ms 转换；低功耗，典型 1 μA，待机小于 0.15 μA；QFN 封装 5.0 × 3.0 × 1.7 mm^3；供电 1.8～3.6 V；内置 24 位 ΔΣ ADC 数字气压传感器；工作范围 10～1200 mbar、-40～+85 °C；支持 I2C 与 SPI（原稿首页标注最高 20 MHz，第 4 页明确 20 MHz 为 SPI 时钟）；无需外部元件（内置振荡器）；长期稳定性优异。

说明

MS5611-01BA 是 MEAS Switzerland 推出的新一代高分辨率高度计传感器，具有 SPI 和 I2C 总线接口，针对高度计和升降率计优化，高度分辨率可达 10 cm。模块由高线性度压力传感器、超低功耗 24 位 ΔΣ ADC 和出厂校准系数组成。它提供 24 位压力、温度数字量，并通过不同工作模式在转换速度与电流消耗之间进行选择。高分辨率温度输出可用于实现高度计兼温度计，无需增加温度传感器。

该器件几乎可连接任何微控制器；通信协议简单，无需配置内部寄存器。5.0 × 3.0 mm、厚 1.7 mm 的尺寸适合移动设备。模块基于 MEMS 技术及 MEAS Switzerland 的批量制造经验；其传感原理使压力和温度信号具有很低的迟滞与较高稳定性。

应用领域

移动高度计/气压计系统；自行车码表；升降率计；医疗报警高度检测；室内导航。

主要技术数据（VDD = 3 V）

项目	数值	单位
压力范围	10～1200	mbar
ADC 位数	24	位
压力 RMS 分辨率，OSR 256/512/1024/2048/4096	0.065/0.042/0.027/0.018/0.012	mbar
25 °C、750 mbar 精度	±1.5	mbar
-20～+85 °C、450～1100 mbar，单点调零后误差	±2.5	mbar
响应时间，OSR 256/512/1024/2048/4096	0.5/1.1/2.1/4.1/8.22	ms
长期稳定性（典型）	-1	mbar/年
温度范围 / 分辨率 / 25 °C 精度	-40～+85 / <0.01 / ±0.8	°C

功能框图：压力敏感元件 → 测量多路器/可编程增益放大器 → ADC/数字滤波 → 数字接口；128 位 PROM 保存校准数据。接口引脚见原稿图。

PERFORMANCE SPECIFICATIONS

ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit
Supply voltage	V_{DD}		-0.3		+4.0	V
Storage temperature	T_S		-40		+125	°C
Overpressure	P_{max}				6	bar
Maximum Soldering Temperature	T_{max}	40 sec max			250	°C
ESD rating		Human Body Model	-4		+4	kV
Latch up		JEDEC standard No 78	-100		+100	mA

ELECTRICAL CHARACTERISTICS

Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit
Operating Supply voltage	V_{DD}		1.8	3.0	3.6	V
Operating Temperature	T		-40	+25	+85	°C
Supply current (1 sample per sec.)	I_{DD}	OSR 4096 2048 1024 512 256		12.5 6.3 3.2 1.7 0.9		µA
Peak supply current		during conversion		1.4		mA
Standby supply current		at 25°C		0.02	0.14	µA
VDD Capacitor		From VDD to GND	100			nF

ANALOG DIGITAL CONVERTER (ADC)

Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit
Output Word				24		bit
Conversion time	t_c	OSR 4096 2048 1024 512 256	7.40 3.72 1.88 0.95 0.48	8.22 4.13 2.08 1.06 0.54	9.04 4.54 2.28 1.17 0.60	ms

性能规格：绝对最大额定值

参数	符号/条件	最小	典型	最大	单位
供电电压	VDD	-0.3		+4.0	V
存储温度	TS	-40		+125	° C
过压	Pmax			6	bar
最高焊接温度	Tmax，最长 40 s			250	° C
ESD 等级	人体模型	-4		+4	kV
闩锁电流	JEDEC 标准 No. 78	-100		+100	mA

电气特性

参数	条件	最小	典型	最大	单位
工作电压 VDD		1.8	3.0	3.6	V
工作温度 T		-40	+25	+85	° C
供电电流 IDD	每秒采样 1 次；OSR 4096/2048/1024/512/256		12.5/6.3/3.2/1.7/0.9		μA
转换期间峰值电流			1.4		mA
待机供电电流	25 ° C		0.02	0.14	μA
VDD 去耦电容	VDD 至 GND	100			nF

模数转换器（ADC）

参数	OSR	最小	典型	最大	单位
输出字长			24		位
转换时间 tc	4096	7.40	8.22	9.04	ms
	2048	3.72	4.13	4.54	ms
	1024	1.88	2.08	2.28	ms
	512	0.95	1.06	1.17	ms
	256	0.48	0.54	0.60	ms

PERFORMANCE SPECIFICATIONS (CONTINUED)

PRESSURE OUTPUT CHARACTERISTICS (V_{DD} = 3 V, T = 25°C UNLESS OTHERWISE NOTED)

Parameter	Conditions		Min.	Typ.	Max	Unit
Operating Pressure Range	P_{range}	Full Accuracy	450		1100	mbar
Extended Pressure Range	P_{ext}	Linear Range of ADC	10		1200	mbar
Total Error Band, no autozero	at 25°C, 700..1100 mbar at 0..50°C, 450..1100 mbar at -20..85°C, 450..1100 mbar at -40..85°C, 450..1100 mbar		-1.5 -2.0 -3.5 -6.0		+1.5 +2.0 +3.5 +6.0	mbar
Total Error Band, autozero at one pressure point	at 25°C, 700..1100 mbar at 10..50°C, 450..1100 mbar at -20..85°C, 450..1100 mbar at -40..85°C, 450..1100 mbar		-0.5 -1.0 -2.5 -5.0		+0.5 +1.0 +2.5 +5.0	mbar
Maximum error with supply voltage	V_{DD} = 1.8 V … 3.6 V		-2.5		+2.5	mbar
Long-term stability				-1		mbar/yr
Recovering time after reflow (1)				7		days
Resolution RMS	OSR	4096 2048 1024 512 256		0.012 0.018 0.027 0.042 0.065		mbar

(1) Time to recovering at least 66% of the reflow impact

TEMPERATURE OUTPUT CHARACTERISTICS (V_{DD} = 3 V, T = 25°C UNLESS OTHERWISE NOTED)

Parameter	Conditions		Min.	Typ.	Max	Unit
Absolute Accuracy	at 25°C -20..85°C -40..85°C		-0.8 -2.0 -4.0		+0.8 +2.0 +4.0	°C
Maximum error with supply voltage	V_{DD} = 1.8 V … 3.6 V		-0.5		+0.5	°C
Resolution RMS	OSR	4096 2048 1024 512 256		0.002 0.003 0.005 0.008 0.012		°C

压力输出特性（除非另有说明，VDD = 3 V，T = 25 ° C）

参数/条件	最小	典型	最大	单位
全精度工作压力范围	450		1100	mbar
扩展压力范围（ADC 线性范围）	10		1200	mbar
总误差带，无单点调零：25 ° C、700～1100 mbar	-1.5		+1.5	mbar
无单点调零：0～50 ° C、450～1100 mbar	-2.0		+2.0	mbar
无单点调零：-20～85 ° C、450～1100 mbar	-3.5		+3.5	mbar
无单点调零：-40～85 ° C、450～1100 mbar	-6.0		+6.0	mbar
单一压力点调零后：25 ° C、700～1100 mbar	-0.5		+0.5	mbar
调零后：10～50 ° C、450～1100 mbar	-1.0		+1.0	mbar
调零后：-20～85 ° C、450～1100 mbar	-2.5		+2.5	mbar
调零后：-40～85 ° C、450～1100 mbar	-5.0		+5.0	mbar
供电变化导致的最大误差，VDD = 1.8～3.6 V	-2.5		+2.5	mbar
长期稳定性		-1		mbar/年
回流焊后恢复时间		7		天

压力 RMS 分辨率：OSR 4096/2048/1024/512/256 时，依次为 0.012/0.018/0.027/0.042/0.065 mbar。回流焊后恢复时间指至少恢复回流焊影响的 66% 所需时间。

温度输出特性（除非另有说明，VDD = 3 V，T = 25 ° C）

绝对精度：25 ° C 为 ±0.8 ° C；-20～85 ° C 为 ±2.0 ° C；-40～85 ° C 为 ±4.0 ° C。VDD 在 1.8～3.6 V 之间变化造成的最大温度误差为 ±0.5 ° C。

温度 RMS 分辨率：OSR 4096/2048/1024/512/256 时，依次为 0.002/0.003/0.005/0.008/0.012 ° C。

PERFORMANCE SPECIFICATIONS (CONTINUED)

DIGITAL INPUTS (CSB, I²C, DIN, SCLK)

Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit
Serial data clock	SCLK	SPI protocol			20	MHz
Input high voltage	V_{IH}	Pins CSB	80% V_{DD}		100% V_{DD}	V
Input low voltage	V_{IL}		0% V_{DD}		20% V_{DD}	V
Input leakage current	$I_{leak25°C}$ $I_{leak85°C}$	at 25°C			0.15	µA

PRESSURE OUTPUTS (I²C, DOUT)

Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit
Output high voltage	V_{OH}	I_{source} = 1.0 mA	80% V_{DD}		100% V_{DD}	V
Output low voltage	V_{OL}	I_{sink} = 1.0 mA	0% V_{DD}		20% V_{DD}	V
Load capacitance	C_{LOAD}				16	pF

性能规格（续）：数字输入

参数	符号/条件	最小	最大	单位
串行数据时钟	SCLK；SPI 协议		20	MHz
输入高电平电压	VIH；CSB 引脚	80% VDD	100% VDD	V
输入低电平电压	VIL	0% VDD	20% VDD	V
输入漏电流	Ileak25° C / Ileak85° C；25 ° C		0.15	μA

输出（原稿标题为 PRESSURE OUTPUTS）

参数	符号/条件	最小	最大	单位
输出高电平电压	VOH；源电流 1.0 mA	80% VDD	100% VDD	V
输出低电平电压	VOL；灌电流 1.0 mA	0% VDD	20% VDD	V
负载电容	CLOAD		16	pF

表内信号名和单位按原稿保留。

FUNCTIONAL DESCRIPTION

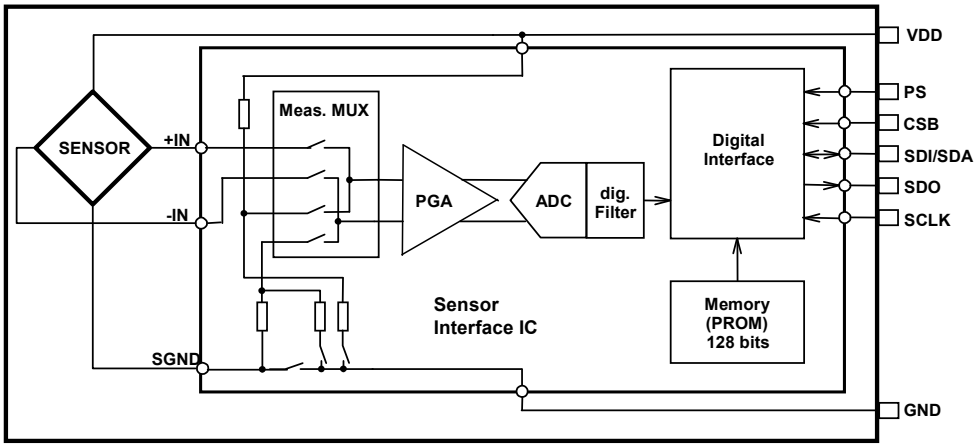


Figure 1: Block diagram of MS5611-01BA

GENERAL

The MS5611-01BA consists of a piezo-resistive sensor and a sensor interface IC. The main function of the MS5611-01BA is to convert the uncompensated analogue output voltage from the piezo-resistive pressure sensor to a 24-bit digital value, as well as providing a 24-bit digital value for the temperature of the sensor.

FACTORY CALIBRATION

Every module is individually factory calibrated at two temperatures and two pressures. As a result, 6 coefficients necessary to compensate for process variations and temperature variations are calculated and stored in the 128-bit PROM of each module. These bits (partitioned into 6 coefficients) must be read by the microcontroller software and used in the program converting D1 and D2 into compensated pressure and temperature values.

SERIAL INTERFACE

The MS5611-01BA has built in two types of serial interfaces: SPI and I^2C. Pulling the Protocol Select pin PS to low selects the SPI protocol, pulling PS to high activates the I^2C bus protocol.

Pin PS	Mode	Pins used
High	I^2C	SDA
Low	SPI	SDI, SDO, CSB

SPI MODE

The external microcontroller clocks in the data through the input SCLK (Serial CLocK) and SDI (Serial Data In). In the SPI mode module can accept both mode 0 and mode 3 for the clock polarity and phase. The sensor responds on the output SDO (Serial Data Out). The pin CSB (Chip Select) is used to enable/disable the interface, so that other devices can talk on the same SPI bus. The CSB pin can be pulled high after the command is sent or after the end of the command execution (for example end of conversion). The best noise performance from the module is obtained when the SPI bus is idle and without communication to other devices during the ADC conversion.

功能说明

图 1：MS5611-01BA 功能框图。SENSOR = 压力敏感元件；Meas. MUX = 测量多路器；PGA = 可编程增益放大器；ADC = 模数转换器；dig. Filter = 数字滤波器；Digital Interface = 数字接口；PROM = 可编程只读存储器。

概述

MS5611-01BA 由压阻式压力传感器和传感器接口 IC 组成。其主要功能是把压阻式压力传感器未经补偿的模拟输出电压转换为 24 位数字量，同时提供传感器温度的 24 位数字量。

出厂校准

每个模块都分别在两个温度点和两个压力点进行出厂校准。由此计算出补偿工艺差异和温度变化所需的 6 个系数，并存入每个模块的 128 位 PROM。微控制器软件必须读取这些位（组成 6 个系数），在程序中把 D1、D2 转换成经过补偿的压力和温度值。

串行接口

器件内置 SPI 和 I2C 两种串行接口。协议选择引脚 PS 接低电平时使用 SPI；接高电平时启用 I2C。

PS 电平	模式	使用的引脚
高	I2C	SDA
低	SPI	SDI、SDO、CSB

SPI 模式

外部微控制器通过 SCLK（串行时钟）和 SDI（串行数据输入）向器件送入数据。SPI 时钟极性与相位可采用模式 0 或模式 3。器件通过 SDO（串行数据输出）应答。CSB（片选）用于使能/禁用接口，以便其他器件共享 SPI 总线。发送命令后或命令执行结束后（例如转换结束）均可把 CSB 拉高。ADC 转换期间让 SPI 总线空闲、且不与其他设备通信，可得到最佳噪声性能。

I²C MODE

The external microcontroller clocks in the data through the input SCLK (Serial CLocK) and SDA (Serial DAta). The sensor responds on the same pin SDA which is bidirectional for the I²C bus interface. So this interface type uses only 2 signal lines and does not require a chip select, which can be favorable to reduce board space. In I²C-Mode the complement of the pin CSB (Chip Select) represents the LSB of the I²C address. It is possible to use two sensors with two different addresses on the I²C bus. The pin CSB shall be connected to VDD or GND (do not leave unconnected!).

COMMANDS

The MS5611-01BA has only five basic commands:

1. Reset
2. Read PROM (128 bit of calibration words)
3. D1 conversion
4. D2 conversion
5. Read ADC result (24 bit pressure / temperature)

I2C 模式

外部微控制器通过时钟线和 SDA（串行数据线）传输数据，器件也通过双向 SDA 应答。该接口只使用两根信号线且不需要片选，可节省电路板空间。在 I2C 模式下，CSB 引脚电平的反值决定 I2C 地址的最低位，因此同一总线上可接两个不同地址的传感器。CSB 必须接 VDD 或 GND，不能悬空。

原稿把 I2C 时钟线写为 SCLK；这里按其功能译为时钟线。

命令

器件只有五类基本命令：1. 复位；2. 读取 PROM（128 位校准字）；3. D1 转换；4. D2 转换；5. 读取 ADC 结果（24 位压力或温度）。

PRESSURE AND TEMPERATURE CALCULATION

Start
Maximum values for calculation results:
P_{MIN} = 10mbar P_{MAX} = 1200mbar
T_{MIN} = -40°C T_{MAX} = 85°C T_{REF} = 20°C

Read calibration data (factory calibrated) from PROM

Variable	Description | Equation	Recommended variable type	Size [1] [bit]	Value min	Value max	Example / Typical
C1	Pressure sensitivity | $SENS_{T1}$	unsigned int 16	16	0	65535	40127
C2	Pressure offset | OFF_{T1}	unsigned int 16	16	0	65535	36924
C3	Temperature coefficient of pressure sensitivity | TCS	unsigned int 16	16	0	65535	23317
C4	Temperature coefficient of pressure offset | TCO	unsigned int 16	16	0	65535	23282
C5	Reference temperature | T_{REF}	unsigned int 16	16	0	65535	33464
C6	Temperature coefficient of the temperature | TEMPSENS	unsigned int 16	16	0	65535	28312

Read digital pressure and temperature data

D1	Digital pressure value	unsigned int 32	24	0	16777216	9085466
D2	Digital temperature value	unsigned int 32	24	0	16777216	8569150

Calculate temperature

dT	Difference between actual and reference temperature [2] $dT = D2 - T_{REF} = D2 - C5 * 2^8$	signed int 32	25	-16776960	16777216	2366
TEMP	Actual temperature (-40…85°C with 0.01°C resolution) $TEMP = 20°C + dT * TEMPSENS = 2000 + dT * C6 / 2^{23}$	signed int 32	41	-4000	8500	2007 = 20.07 °C

Calculate temperature compensated pressure

OFF	Offset at actual temperature [3] $OFF = OFF_{T1} + TCO * dT = C2 * 2^{16} + (C4 * dT) / 2^7$	signed int 64	41	-8589672450	12884705280	2420281617
SENS	Sensitivity at actual temperature [4] $SENS = SENS_{T1} + TCS * dT = C1 * 2^{15} + (C3 * dT) / 2^8$	signed int 64	41	-4294836225	6442352640	1315097036
P	Temperature compensated pressure (10…1200mbar with 0.01mbar resolution) $P = D1 * SENS - OFF = (D1 * SENS / 2^{21} - OFF) / 2^{15}$	signed int 32	58	1000	120000	100009 = 1000.09 mbar

Display pressure and temperature value

Notes
[1] Maximal size of intermediate result during evaluation of variable
[2] min and max have to be defined
[3] min and max have to be defined
[4] min and max have to be defined

Figure 2: Flow chart for pressure and temperature reading and software compensation.

压力与温度计算

计算范围：压力 P = 10～1200 mbar；温度 T = -40～85 ° C；参考温度 TREF = 20 ° C。先读取出厂校准 PROM，再读取 D1（压力）和 D2（温度）原始数字值。

变量	含义	推荐类型	原稿典型值
C1	压力灵敏度 SENST1	无符号 16 位	40127
C2	压力偏移 OFFT1	无符号 16 位	36924
C3	压力灵敏度温度系数 TCS	无符号 16 位	23317
C4	压力偏移温度系数 TCO	无符号 16 位	23282
C5	参考温度 TREF	无符号 16 位	33464
C6	温度系数 TEMPSENS	无符号 16 位	28312
D1	压力原始数字量（24 位）	无符号 32 位	9085466
D2	温度原始数字量（24 位）	无符号 32 位	8569150

一阶补偿公式

dT = D2 - C5 × 2^8 = 2366。

TEMP = 2000 + dT × C6 / 2^23 = 2007，即 20.07 ° C。

OFF = C2 × 2^16 + (C4 × dT) / 2^7 = 2420281617。

SENS = C1 × 2^15 + (C3 × dT) / 2^8 = 1315097036。

P = (D1 × SENS / 2^21 - OFF) / 2^15 = 100009，即 1000.09 mbar。

TEMP 以 0.01 ° C 为单位，P 以 0.01 mbar 为单位。dT 推荐有符号 32 位；OFF、SENS 推荐有符号 64 位；TEMP、P 推荐有符号 32 位。乘法和中间结果也须具有足够位宽，避免溢出。

图 2：读取校准系数 → 读取 D1、D2 → 计算温度 → 计算温度补偿压力 → 显示压力与温度。原稿表格中的部分位宽/范围栏排版存在可疑之处，以上公式和典型值按图中内容保留。

SECOND ORDER TEMPERATURE COMPENSATION

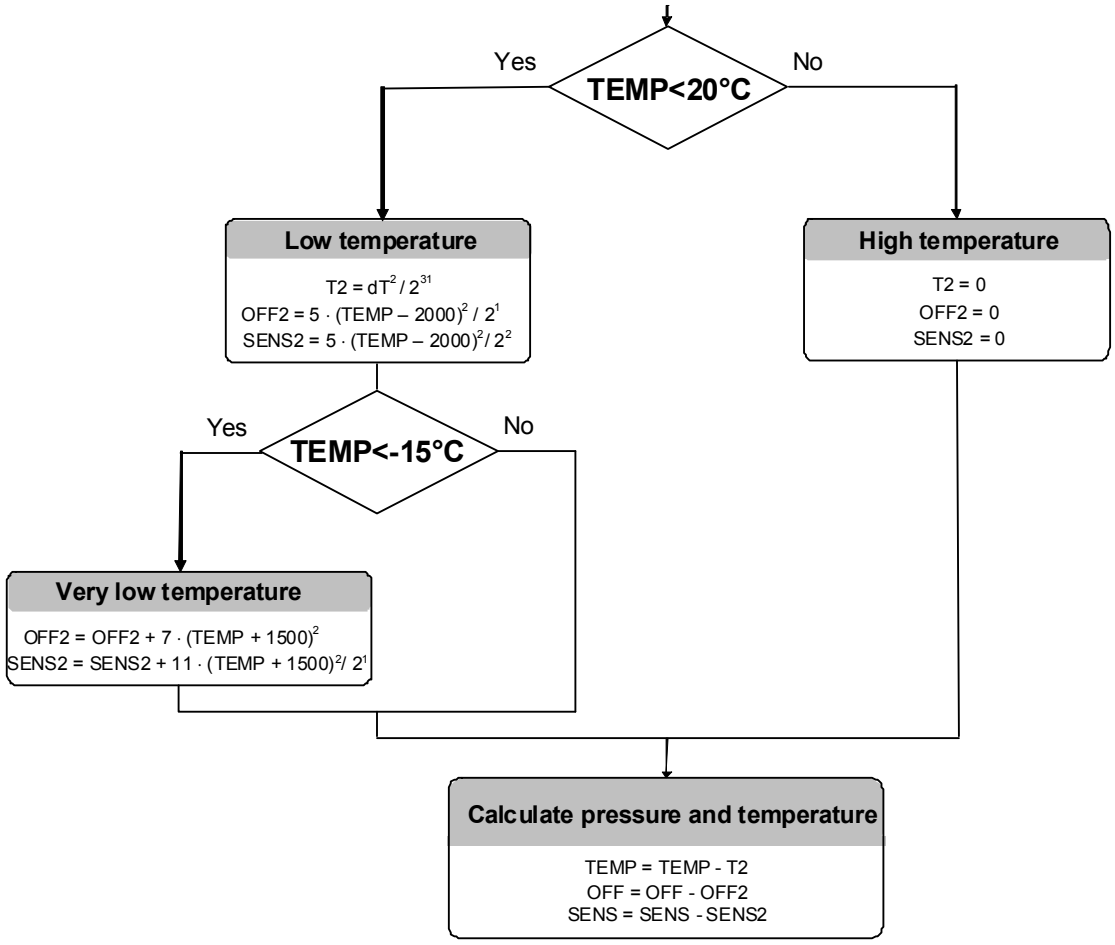


Figure 3: Flow chart for pressure and temperature to the optimum accuracy.

二阶温度补偿

先用第 7 页公式求得一阶 TEMP、OFF 和 SENS，再按一阶 TEMP 判断低温分支。这里 TEMP 的单位仍为 0.01 °C。

当 TEMP < 2000（即低于 20 °C）

T2 = dT^2 / 2^31

OFF2 = 5 × (TEMP - 2000)^2 / 2^1

SENS2 = 5 × (TEMP - 2000)^2 / 2^2

当 TEMP < -1500（即低于 -15 °C），再追加

OFF2 = OFF2 + 7 × (TEMP + 1500)^2

SENS2 = SENS2 + 11 × (TEMP + 1500)^2 / 2^1

当 TEMP ≥ 2000

T2 = 0；OFF2 = 0；SENS2 = 0。

修正结果

TEMP = TEMP - T2

OFF = OFF - OFF2

SENS = SENS - SENS2

最后用修正后的 OFF 与 SENS 重算压力 P。

图 3：为达到最佳精度而进行的压力与温度二阶补偿流程。原图的判断框写作 TEMP<20° C、TEMP<-15° C；实施公式时对应 2000 和 -1500。

SPI INTERFACE

COMMANDS

Size of each command is 1 byte (8 bits) as described in the table below. After ADC read commands the device will return 24 bit result and after the PROM read 16bit result. The address of the PROM is embedded inside of the PROM read command using the a2, a1 and a0 bits.

	Command byte								hex value
Bit number	0	1	2	3	4	5	6	7	
Bit name	PRM	COV	-	Typ	Ad2/ Os2	Ad1/ Os1	Ad0/ Os0	Stop	
Command									
Reset	0	0	0	1	1	1	1	0	0x1E
Convert D1 (OSR=256)	0	1	0	0	0	0	0	0	0x40
Convert D1 (OSR=512)	0	1	0	0	0	0	1	0	0x42
Convert D1 (OSR=1024)	0	1	0	0	0	1	0	0	0x44
Convert D1 (OSR=2048)	0	1	0	0	0	1	1	0	0x46
Convert D1 (OSR=4096)	0	1	0	0	1	0	0	0	0x48
Convert D2 (OSR=256)	0	1	0	1	0	0	0	0	0x50
Convert D2 (OSR=512)	0	1	0	1	0	0	1	0	0x52
Convert D2 (OSR=1024)	0	1	0	1	0	1	0	0	0x54
Convert D2 (OSR=2048)	0	1	0	1	0	1	1	0	0x56
Convert D2 (OSR=4096)	0	1	0	1	1	0	0	0	0x58
ADC Read	0	0	0	0	0	0	0	0	0x00
PROM Read	1	0	1	0	Ad2	Ad1	Ad0	0	0xA0 to 0xAE

Figure 4: Command structure

RESET SEQUENCE

The Reset sequence shall be sent once after power-on to make sure that the calibration PROM gets loaded into the internal register. It can be also used to reset the device ROM from an unknown condition

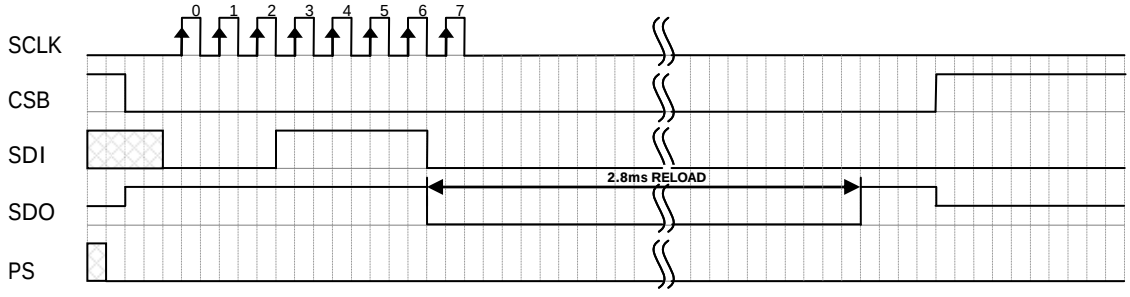


Figure 5: Reset command sequence SPI mode 0

SPI 接口：命令

每条命令长 1 字节（8 位）。ADC 读取返回 24 位结果；PROM 读取返回 16 位结果。PROM 地址由读取命令中的 a2、a1、a0 位编码。

命令	十六进制
复位	0x1E
转换 D1，OSR=256/512/1024/2048/4096	0x40/0x42/0x44/0x46/0x48
转换 D2，OSR=256/512/1024/2048/4096	0x50/0x52/0x54/0x56/0x58
读取 ADC	0x00
读取 PROM，地址 0～7	0xA0～0xAE，步长 2

图 4：命令位结构。原稿从 bit 0 到 bit 7 列出 PRM、COV、Typ、Ad2/Os2、Ad1/Os1、Ad0/Os0 与 Stop 等字段；上表给出实际命令字节。

SPI 复位时序

上电后应发送一次复位命令，使校准 PROM 装入内部寄存器。器件状态不明时也可用复位命令恢复。原稿图 5 为 SPI 模式 0 的复位命令时序，图 6 为模式 3；两图均标出约 2.8 ms 的重新装载时间。

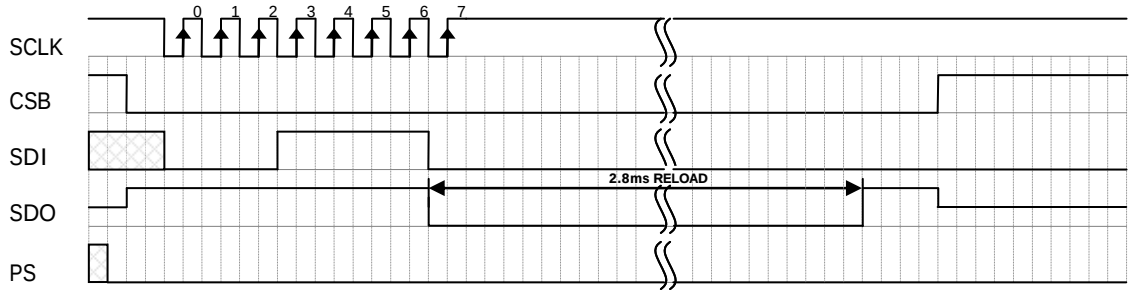


Figure 6: Reset command sequence SPI mode 3

CONVERSION SEQUENCE

The conversion command is used to initiate uncompensated pressure (D1) or uncompensated temperature (D2) conversion. The chip select can be disabled during this time to communicate with other devices.
After the conversion, using ADC read command the result is clocked out with the MSB first. If the conversion is not executed before the ADC read command, or the ADC read command is repeated, it will give 0 as the output result. If the ADC read command is sent during conversion the result will be 0, the conversion will not stop and the final result will be wrong. Conversion sequence sent during the already started conversion process will yield incorrect result as well.

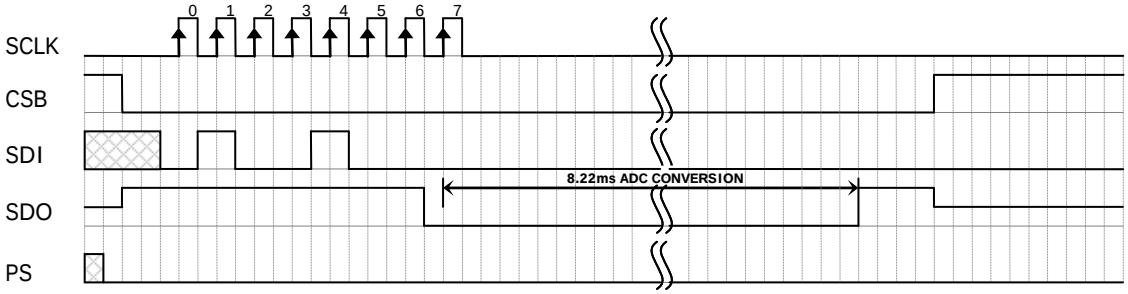


Figure 7: Conversion out sequence, Typ=d1, OSR = 4096

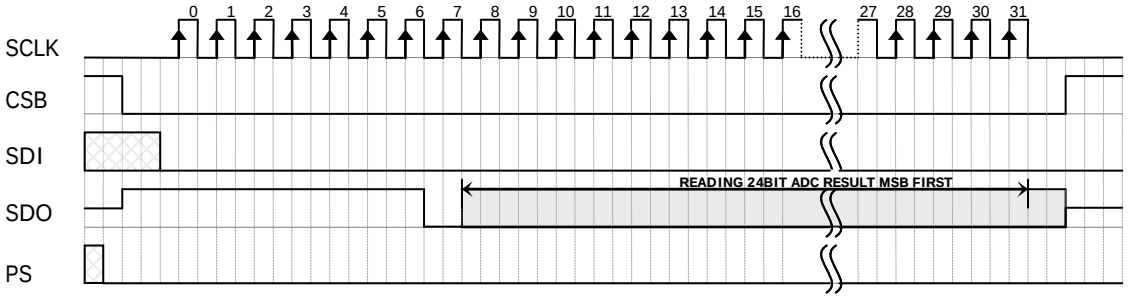


Figure 8: ADC Read sequence

PROM READ SEQUENCE

The read command for PROM shall be executed once after reset by the user to read the content of the calibration PROM and to calculate the calibration coefficients. There are in total 8 addresses resulting in a total memory of 128 bit. Address 0 contains factory data and the setup, addresses 1-6 calibration coefficients and address 7 contains the serial code and CRC. The command sequence is 8 bits long with a 16 bit result which is clocked with the MSB first.

SPI 接口：转换、ADC 读取与 PROM 读取

转换序列

发送转换命令以开始未补偿压力 D1 或未补偿温度 D2 的转换。转换期间可以取消片选以便与其他器件通信。转换结束后发送 ADC 读取命令，结果按最高有效位先出。

若未执行转换便读取 ADC，或重复读取 ADC，输出为 0。若在转换期间发送 ADC 读取命令，返回值为 0；转换不会停止，但最终结果会错误。已开始转换时再发送转换命令，也会得到错误结果。

图 7：转换命令时序，示例为 D1、OSR=4096，典型转换时间 8.22 ms。图 8：24 位 ADC 结果读取时序，最高有效位先出。

PROM 读取序列

复位后，用户应读取校准 PROM，用其内容计算补偿系数。PROM 有 8 个地址，共 128 位。地址 0 为出厂数据与设置；地址 1～6 为校准系数；地址 7 为序列码与 CRC。命令长 8 位，读取结果长 16 位，最高有效位先出。

图 9（续至下一原稿页）：地址 011（系数 3）的 PROM 读取时序。

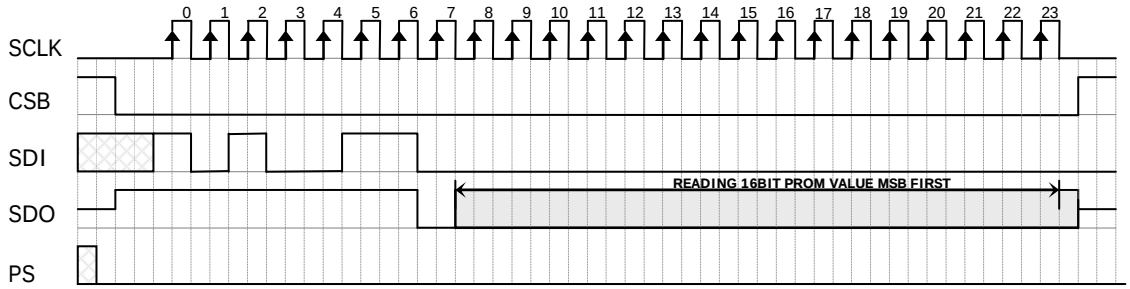


Figure 9: PROM Read sequence, address = 011 (Coefficient 3).

I²C INTERFACE

COMMANDS

Each I²C communication message starts with the start condition and it is ended with the stop condition. The MS5611-01BA address is 111011Cx, where C is the complementary value of the pin CSB. Since the IC does not have a microcontroller inside, the commands for I²C and SPI are quite similar.

RESET SEQUENCE

The reset can be sent at any time. In the event that there is not a successful power on reset this may be caused by the SDA being blocked by the module in the acknowledge state. The only way to get the MS5611-01BA to function is to send several SCLKs followed by a reset sequence or to repeat power on reset.

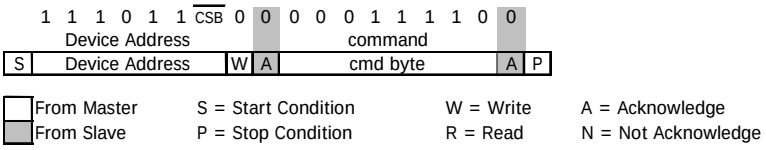


Figure 10: I²C Reset Command

PROM READ SEQUENCE

The PROM Read command consists of two parts. First command sets up the system into PROM read mode. The second part gets the data from the system.

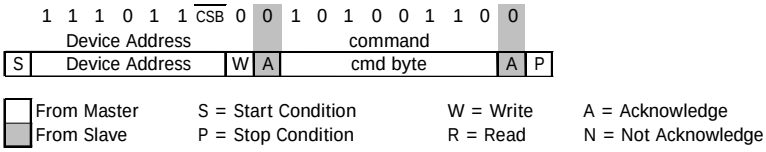


Figure 11: I²C Command to read memory address= 011 (Coefficient 3)

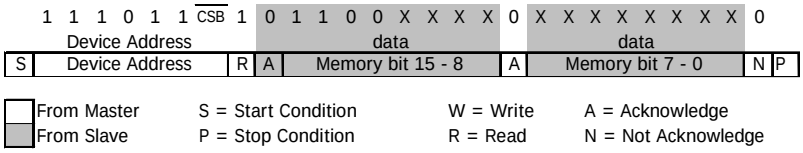


Figure 12: I²C answer from MS5611-01BA

SPI PROM 读取（续）与 I2C 接口

图 9：读取地址 011（系数 3）的 16 位 PROM 值，最高有效位先出。

I2C 命令

每次 I2C 通信以起始条件开始、以停止条件结束。原稿给出的器件地址位串为 111011Cx，其中 C 为 CSB 引脚电平的反值。该器件内部不含微控制器，因此 I2C 与 SPI 的命令很相似。

复位序列

复位可随时发送。若上电复位未成功，模块可能在应答状态占用
SDA；可发送多个时钟脉冲后再执行复位，或重新上电复位。

图 10：I2C 复位命令。S=起始，P=停止，W=写，R=读，A=应答，N=非应答；图中还区分主机与从机发送的位。

PROM 读取序列

PROM 读取分两部分：先发命令使器件进入 PROM 读取模式，再从器件取回数据。

图 11：读取存储器地址 011（系数 3）的 I2C 命令。图 12：器件应答，按高字节（位 15～8）、低字节（位 7～0）返回 PROM 数据。

CONVERSION SEQUENCE

A conversion can be started by sending the command to MS5611-01BA. When command is sent to the system it stays busy until conversion is done. When conversion is finished the data can be accessed by sending a Read command, when an acknowledge appears from the MS5611-01BA, 24 SCLK cycles may be sent to receive all result bits. Every 8 bit the system waits for an acknowledge signal.

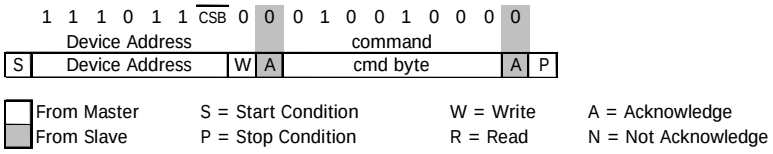


Figure 13: I²C Command to initiate a pressure conversion (OSR=4096, typ=D1)

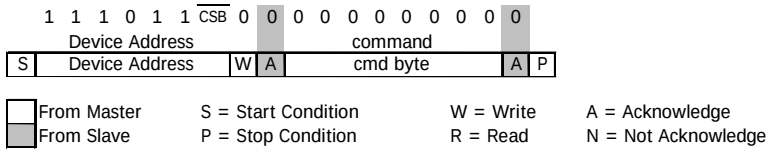


Figure 14: I²C ADC read sequence

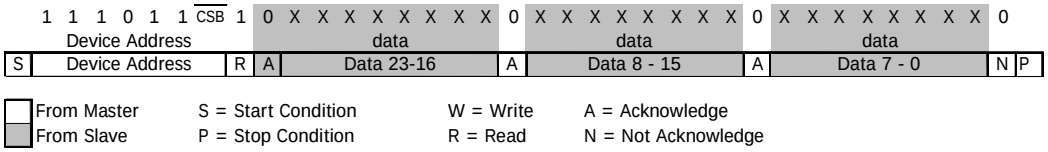


Figure 15: I²C answer from MS5611-01BA

CYCLIC REDUNDANCY CHECK (CRC)

MS5611-01BA contains a PROM memory with 128-Bit. A 4-bit CRC has been implemented to check the data validity in memory. The application note AN520 describes in detail CRC-4 code used.

Add	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
0	16 bit reserved for manufacturer															
1	Coefficient 1 (16 bit unsigned)															
2	Coefficient 2 (16 bit unsigned)															
3	Coefficient 3 (16 bit unsigned)															
4	Coefficient 4 (16 bit unsigned)															
5	Coefficient 5 (16 bit unsigned)															
6	Coefficient 6 (16 bit unsigned)															
7														CRC		

Figure 16: Memory PROM mapping

I2C 转换序列

向 MS5611-01BA 发送相应命令即可启动转换；转换完成前器件保持忙状态。转换结束后发送 ADC 读取命令。收到器件应答后，可产生 24 个时钟周期，接收全部结果位。每接收 8 位，器件等待一个应答信号。

图 13：启动压力 D1 转换（OSR=4096）的 I2C 命令，命令字节 0x48。图 14：ADC 读取命令，字节 0x00。图 15：器件返回 24 位结果，依次为位 23～16、15～8、7～0；最后一字节后主机发送非应答并停止。

循环冗余校验（CRC）

器件有 128 位 PROM。其内实现 4 位 CRC，用于检查存储数据的有效性。CRC-4 算法详见应用说明 AN520。

PROM 地址	内容
0	16 位制造商保留数据
1～6	分别为无符号 16 位校准系数 C1～C6
7	序列信息与 CRC

图 16：PROM 存储映射。原稿图中用 DB15～DB0 表示每个字的位位置。

APPLICATION CIRCUIT

The MS5611-01BA is a circuit that can be used in conjunction with a microcontroller in mobile altimeter applications. It is designed for low-voltage systems with a supply voltage of 3 V.

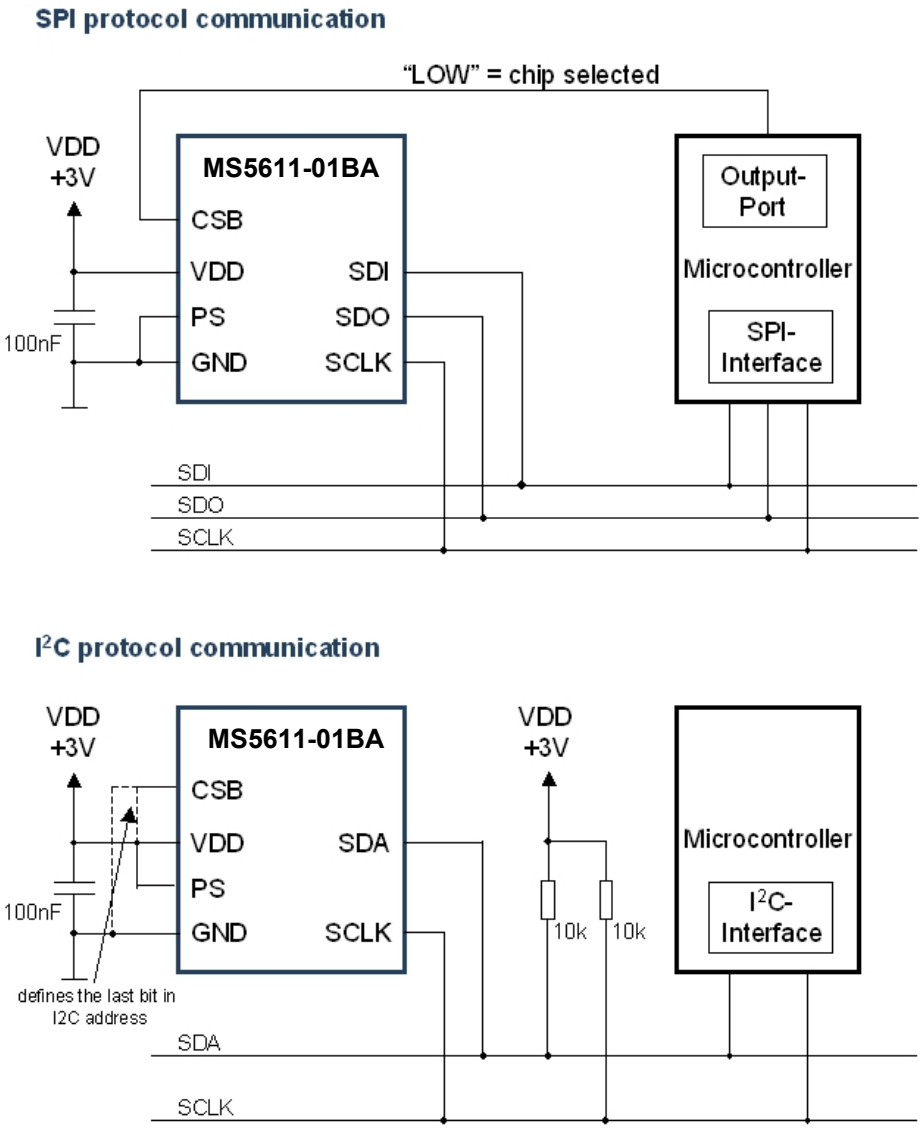


Figure 17: Typical application circuit with SPI / I^2C protocol communication

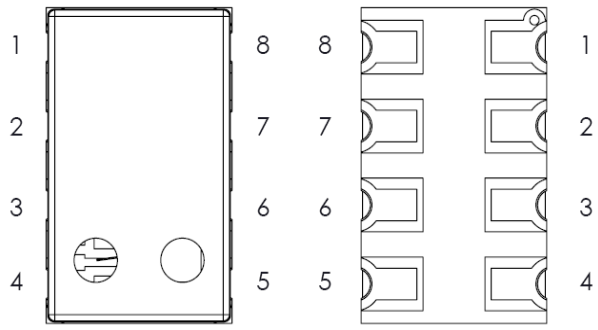
应用电路

MS5611-01BA 可与微控制器配合用于移动高度计。器件针对 3 V 供电的低压系统设计。原稿图 17 给出 SPI 与 I2C 通信的典型应用电路。

图 17：典型应用连接。PS 选择协议；I2C 模式使用 SDA 与时钟线，SPI 模式使用 SDI、SDO、SCLK、CSB。VDD 旁应放置 100 nF 去耦电容，详见第 16 页译文。电路连线及元件位置以对照的原图为准。

PIN CONFIGURATION

Pin	Name	Type	Function
1	VDD	P	Positive supply voltage
2	PS	I	Protocol select PS high (VDD) → I^2C PS low (GND) → SPI
3	GND	G	Ground
4	CSB	I	Chip select (active low), internal connection
5			
6	SDO	O	Serial data output
7	SDI / SDA	I / IO	Serial data input / I^2C data IO
8	SCLK	I	Serial data clock



DEVICE PACKAGE OUTLINE

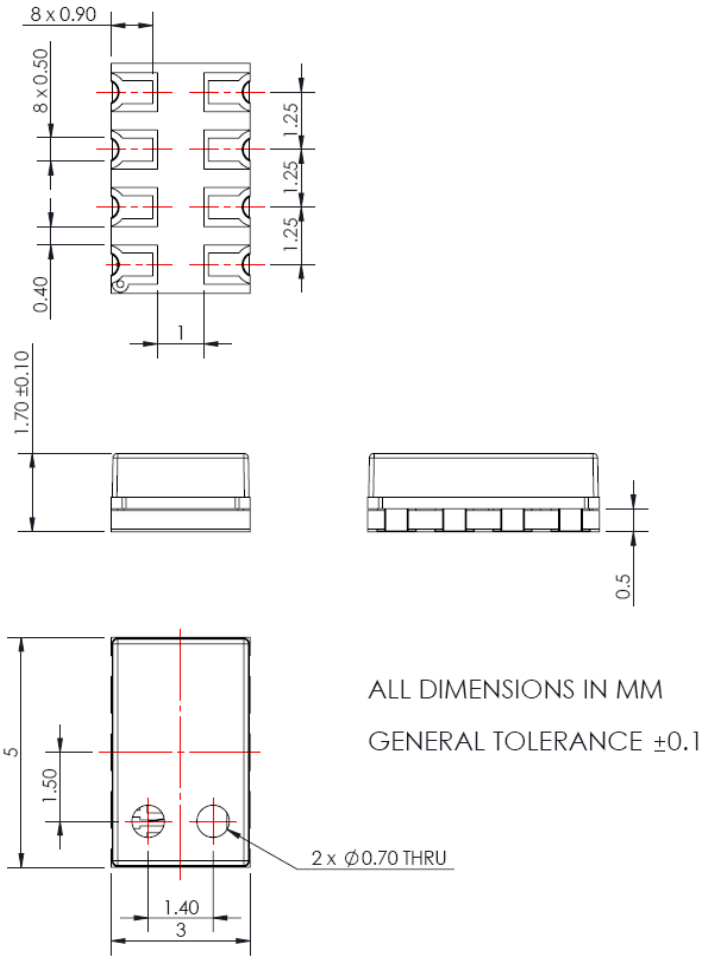


Figure 18: MS5611-01BA01 package outline

引脚配置

引脚	名称	类型	功能
1	VDD	电源	正电源
2	PS	输入	协议选择：接高电平 VDD → I2C；接低电平 GND → SPI
3	GND	地	接地
4、5	CSB	输入	片选，低电平有效；原稿标注内部相连
6	SDO	输出	串行数据输出
7	SDI/SDA	输入/双向	SPI 串行数据输入 / I2C 双向数据
8	SCLK	输入	串行数据时钟

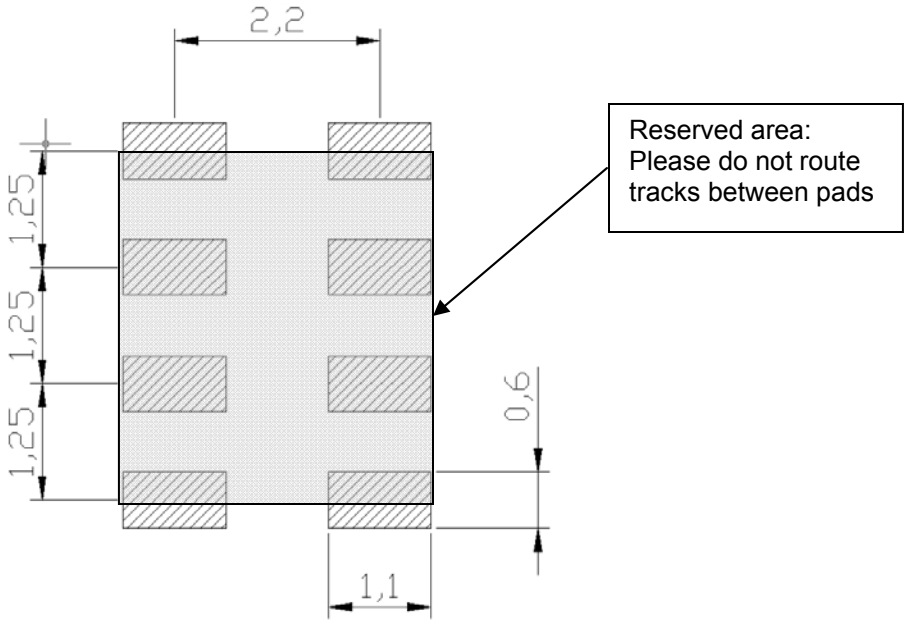
器件封装外形

图 18：MS5611-01BA01

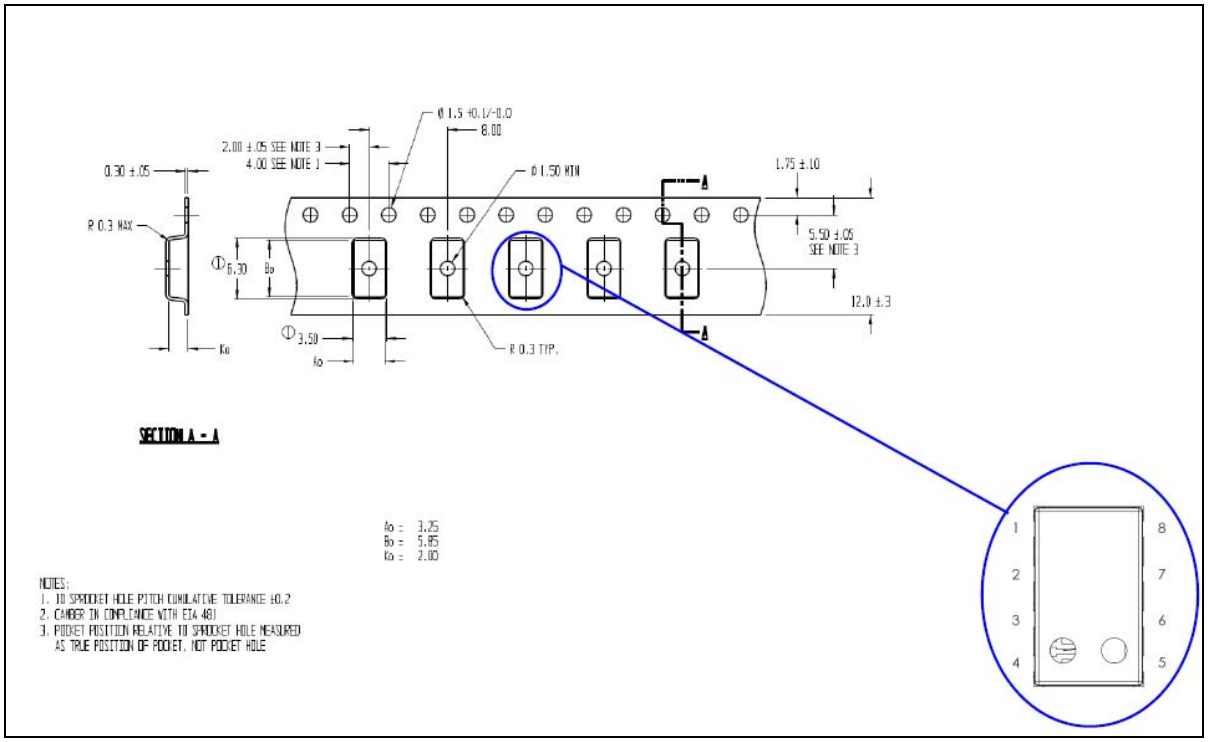
封装外形与尺寸。机械尺寸、焊盘方位和公差请直接按对照的原始工程图读取，勿以本页文字替代尺寸图。

RECOMMENDED PAD LAYOUT

Pad layout for bottom side of the MS5611-01BA soldered onto printed circuit board.



SHIPPING PACKAGE



Tape and reel

推荐焊盘布局

原稿给出 MS5611-01BA 焊接到印制电路板时的底面焊盘布局。图中的保留区域不得在焊盘之间布线。焊盘尺寸与间距请按对照的原图制作。

出货包装

卷带包装（Tape and reel）。

MOUNTING AND ASSEMBLY CONSIDERATIONS

SOLDERING

Please refer to the application note AN808 available on our website for all soldering issues.

MOUNTING

The MS5611-01BA can be placed with automatic Pick & Place equipment using vacuum nozzles. It will not be damaged by the vacuum. Due to the low stress assembly the sensor does not show pressure hysteresis effects. It is important to solder all contact pads.

CONNECTION TO PCB

The package outline of the module allows the use of a flexible PCB for interconnection. This can be important for applications in watches and other special devices.

CLEANING

The MS5611-01BA has been manufactured under cleanroom conditions. It is therefore recommended to assemble the sensor under class 10'000 or better conditions. Should this not be possible, it is recommended to protect the sensor opening during assembly from entering particles and dust. To avoid cleaning of the PCB, solder paste of type "no-clean" shall be used. Cleaning might damage the sensor!

ESD PRECAUTIONS

The electrical contact pads are protected against ESD up to 4 kV HBM (human body model). It is therefore essential to ground machines and personnel properly during assembly and handling of the device. The MS5611-01BA is shipped in antistatic transport boxes. Any test adapters or production transport boxes used during the assembly of the sensor shall be of an equivalent antistatic material.

DECOUPLING CAPACITOR

Particular care must be taken when connecting the device to the power supply. A 100 nF ceramic capacitor must be placed as close as possible to the MS5611-01BA VDD pin. This capacitor will stabilize the power supply during data conversion and thus, provide the highest possible accuracy.

安装与装配注意事项

焊接

所有焊接相关问题请参阅网站上的应用说明 AN808。

贴装

可使用真空吸嘴和自动贴片设备放置 MS5611-01BA；真空不会损伤器件。由于装配应力较低，传感器不会表现出压力迟滞效应。务必焊接所有接触焊盘。

与 PCB 的连接

模块的封装外形允许使用柔性 PCB 进行连接，这对于手表和其他特殊设备可能很重要。

清洁

器件在洁净室条件下制造，因此建议在 10,000 级或更洁净的环境中装配。若无法做到，应在装配期间保护传感器开口，防止颗粒和灰尘进入。应使用免清洗焊膏，以避免清洗 PCB；清洗可能损坏传感器。

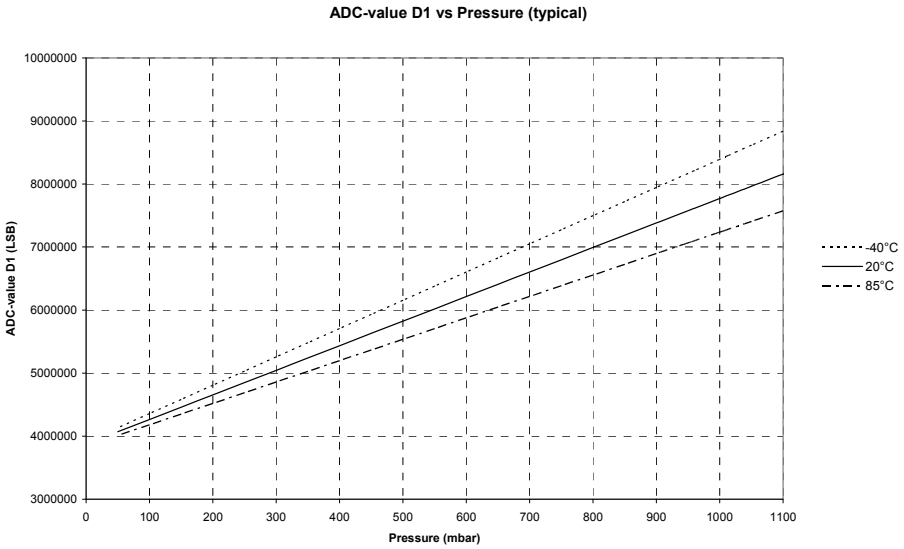
ESD 防护

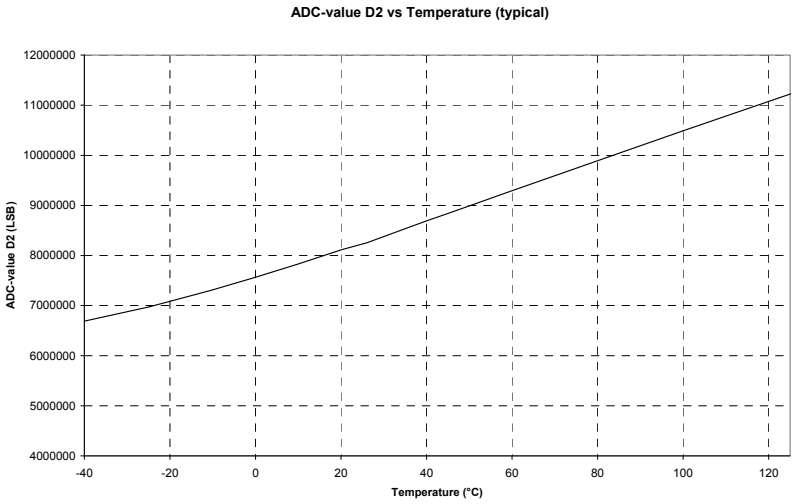
电气接触焊盘可承受人体模型 4 kV 的静电放电。装配和操作期间必须让机器和人员妥善接地。器件以防静电运输盒交付，测试夹具或生产运输盒也应采用等效的防静电材料。

去耦电容

连接电源时应特别注意：在 VDD 引脚附近尽可能靠近地放置 100 nF
陶瓷电容。此电容在数据转换期间稳定电源，以获得尽可能高的精度。

TYPICAL PERFORMANCE CHARACTERISTICS





典型性能曲线

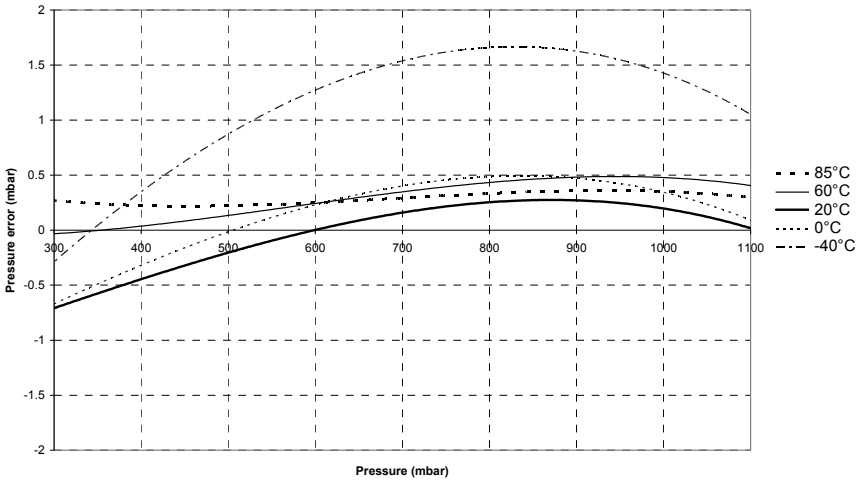
上图：ADC 原始压力值 D1 与压力的关系（典型值）。横轴为压力，单位 mbar；纵轴为 D1 数字量，单位 LSB；曲线对应 −40 °C、20 °C、85 °C。

下图：ADC 原始温度值 D2 与温度的关系（典型值）。横轴为温度，单位 °C；纵轴为 D2 数字量，单位 LSB。

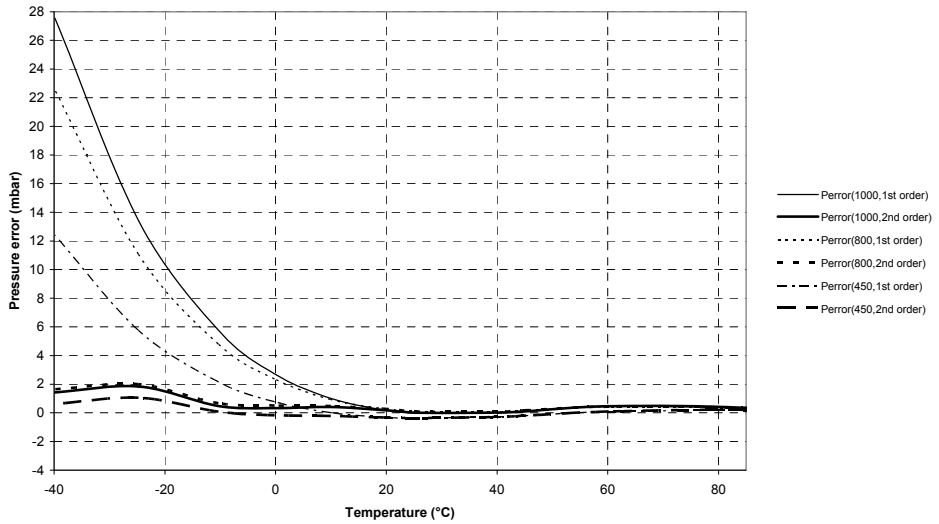
原稿第 17 页保留了曲线和刻度，请用其读取趋势与近似值。

TYPICAL PERFORMANCE CHARACTERISTICS (CONTINUED)

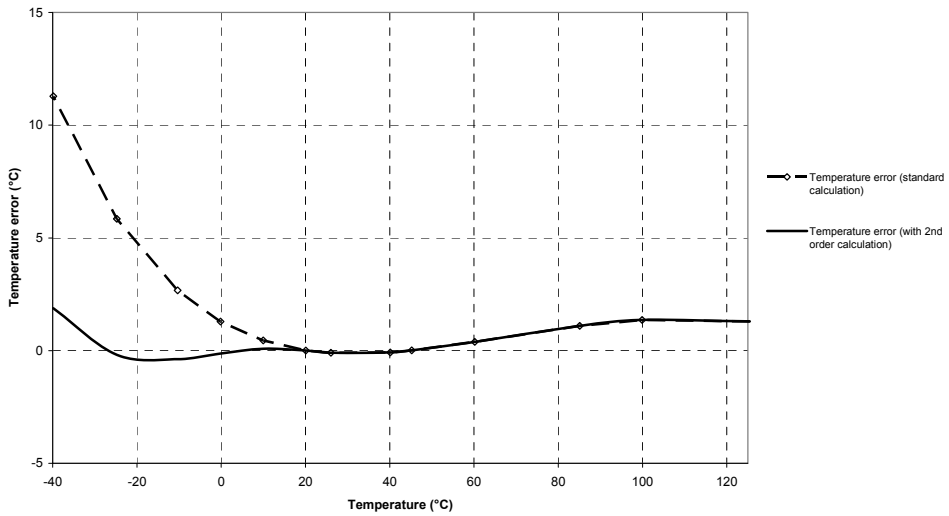
Absolute Pressure Accuracy after Calibration, 2nd order compensation



Pressure Error Accuracy vs temperature (typical)



Temperature Error Accuracy vs temperature (typical)



典型性能曲线（续）

上图：校准并采用二阶补偿后的绝对压力精度。横轴压力（mbar），纵轴压力误差（mbar）；曲线温度为 -40、0、20、60、85 °C，图中 Poly. 表示多项式拟合。

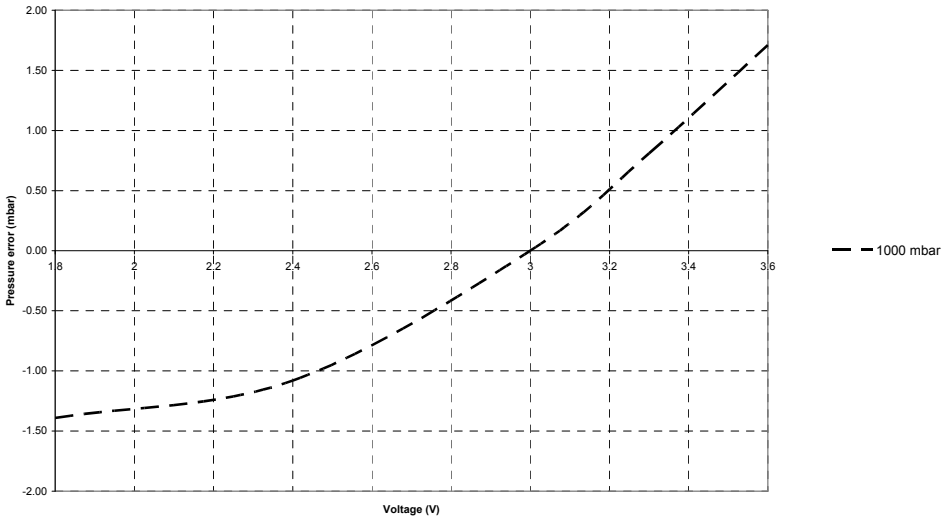
中图：典型压力误差与温度的关系。横轴温度（°C），纵轴压力误差（mbar）；分别显示 450、800、1000 mbar 条件下一阶补偿与二阶补偿的误差曲线。

下图：典型温度误差与温度的关系。横轴温度（°C），纵轴温度误差（°C）；比较标准计算（一阶）与二阶补偿计算。

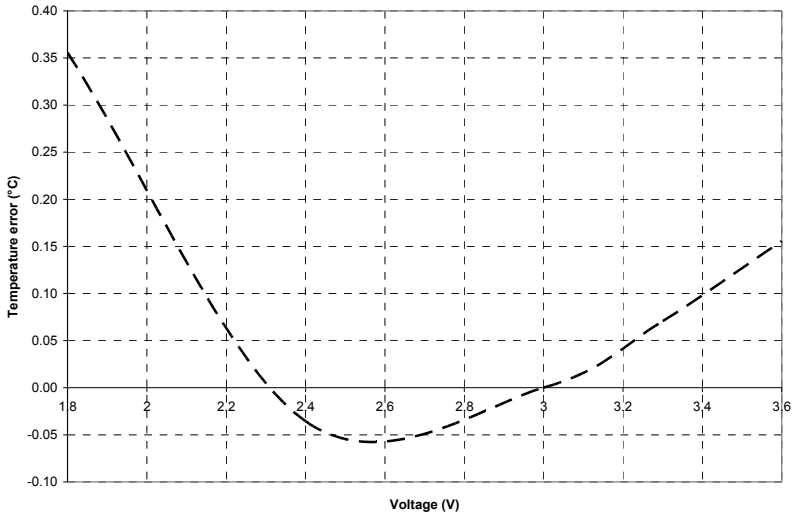
原稿第 18 页保留完整曲线、坐标刻度和图例。

TYPICAL PERFORMANCE CHARACTERISTICS (CONTINUED)

Pressure error vs supply voltage (typical)



Temperature error vs supply voltage (typical)



典型性能曲线（续）

上图：典型温度误差与供电电压的关系。横轴电压 1.8～3.6 V，纵轴温度误差（° C）。

下图：典型压力误差与供电电压的关系。横轴电压 1.8～3.6 V，纵轴压力误差（mbar）；图中标注 1000 mbar。

原稿第 19 页保留完整曲线和坐标刻度。

MS5611-01BA01 Variometer Module, with LCP cap

ORDERING INFORMATION

Product Code	Product	Art. No	Delivery Form
MS5611-01BA01	Variometer Module Plastic Cap	MS561101BA01-50	Tape and reel

NORTH AMERICA

Measurement Specialties
45738 Northport Loop West
Fremont, CA 94538

Tel: +1 800 767 1888
Fax: +1 510 498 1578
e-mail: pfg.cs.amer@meas-spec.com
Website: www.meas-spec.com

EUROPE

MEAS Switzerland Sàrl
Ch. Chapons-des-Prés 11
CH-2022 Bevaix

Tel: +41 32 847 9550
Fax: + 41 32 847 9569
e-mail: sales.ch@meas-spec.com
Website: www.meas-spec.com

ASIA

Measurement Specialties (China), Ltd.
No. 26 Langshan Road
Shenzhen High-Tech Park (North)
Nanshan District, Shenzhen, 518057
China
Tel: +86 755 3330 5088
Fax: +86 755 3330 5099
e-mail: pfg.cs.asia@meas-spec.com
Website: www.meas-spec.com

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订货信息

产品	订货号	交付形式
MS5611-01BA01 升降率计模块，塑料盖	MS561101BA01-50	卷带包装

联系信息

北美：Measurement Specialties，45738 Northport Loop West，Fremont，CA 94538；电话 +1 800 767 1888，传真 +1 510 498 1578；邮箱 pfg.cs.amer@meas-spec.com。

欧洲：MEAS Switzerland Sàrl，Ch. Chapons-des-Prés 11，CH-2022 Bevaix；电话 +41 32 847 9550，传真 +41 32 847 9569；邮箱 sales.ch@meas-spec.com。

亚洲：Measurement Specialties (China), Ltd.，中国深圳市南山区深圳高新技术产业园（北区）朗山路 26 号，邮编 518057；电话 +86 755 3330 5088，传真 +86 755 3330 5099；邮箱 pfg.cs.asia@meas-spec.com。网站：www.meas-spec.com。

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